

Question 1

Let $x \in \mathbb{R}^n$. Its p-norm is defined as

$$\|x\|_p = \left[\sum_{i=1}^n |x_i|^p \right]^{1/p}$$

Prove it is a norm when $p \geq 1$.

proof: 1) Non-negativity and definiteness

Since $|x_i|^p \geq 0$, we have $\sum_{i=1}^n |x_i|^p \geq 0$, hence $\|x\|_p \geq 0$

if $\|x\|_p = 0$, then $\left(\sum_{i=1}^n |x_i|^p \right)^{1/p} = 0 \Rightarrow \sum_{i=1}^n |x_i|^p = 0$

$\Rightarrow$ in this case, for $\forall i$, $x_i = 0 \Rightarrow x = 0$

Conversely if $x = 0$, then $\|x\|_p = 0 \Rightarrow \|x\|_p = 0$ iff. $x = 0$

2) Scaling

For any scalar α , $\|\alpha x\|_p = \left(\sum_{i=1}^n |\alpha x_i|^p \right)^{1/p} = |\alpha| \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$

3) triangular inequality:

Let lemma be: If $\|x\|_p = 1$, $\|y\|_p = 1$, $\forall \theta \in (0, 1)$ then $\|\theta x + (1-\theta)y\|_p \leq 1$

when $\|x\|_p = \|y\|_p = 1$, there is: $\theta \|x\|_p + (1-\theta) \|y\|_p = 1$ ①

Let $f(x) = \|x\|_p^p = \sum_{i=1}^n |x_i|^p$: ②

using second order derivative, we can conclude that $f(x)$ is convex.

$\Rightarrow$ then there is $f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y) = 1$

$\Rightarrow \|\theta x + (1-\theta)y\|_p \leq 1$ is proved.

Now, regarding $\|x\|_p + \|y\|_p$ & $\|x+y\|_p$.

① $x=0$ or $y=0 \Rightarrow$ obviously true.

② x, y both non-zero.

$$\text{let } a = \|x\|_p + \|y\|_p, \quad x_1 = \frac{x}{\|x\|_p}, \quad y_1 = \frac{y}{\|y\|_p}$$

$$\Rightarrow \|x_1\|_p = \|y_1\|_p = 1$$

let $\theta = \frac{\|x\|_p}{a}$ based on the above lemma:

$$\text{there is } \left\| \frac{\|x\|_p}{a} \cdot \frac{x}{\|x\|_p} + \frac{\|y\|_p}{a} \cdot \frac{y}{\|y\|_p} \right\| = \left\| \frac{x+y}{\|x\|_p + \|y\|_p} \right\|_p \leq 1$$

$$\Rightarrow \|x+y\|_p \leq \|x\|_p + \|y\|_p \quad \text{Done proving.}$$

Question 2

Let $p(x)$ and $q(x)$ are two probability distributions of a discrete random variable x on X . That is, both $p(x)$ and $q(x)$ sum up to 1, and $p(x) > 0$ and $q(x) > 0$. The Kullback-Leibler divergence is defined as

$$DKL(p(x), q(x)) = \sum_{x \in X} p(x) \log\left(\frac{p(x)}{q(x)}\right) \geq 0$$

Prove

$$DKL(p(x), q(x)) \geq 0.$$

$$\text{proof: let } t = \frac{q(x)}{p(x)} > 0, \text{ then } -\log\left(\frac{q(x)}{p(x)}\right) \geq 1 - \frac{q(x)}{p(x)}$$

$$\Rightarrow p(x) \cdot \left[-\log\left(\frac{q(x)}{p(x)}\right)\right] \geq p(x) - q(x)$$

$$\text{since } p(x) \cdot \left[-\log\left(\frac{q(x)}{p(x)}\right)\right] = p(x) \cdot \log\left(\frac{p(x)}{q(x)}\right)$$

$$\Rightarrow p(x) \log\left(\frac{p(x)}{q(x)}\right) \geq p(x) - q(x)$$

$\Rightarrow$ for all $x \in X$,

$$\sum_{x \in X} p(x) \log\left(\frac{p(x)}{q(x)}\right) \geq \sum_{x \in X} [p(x) - q(x)] = 1 - 1 = 0$$

$$\Rightarrow D_{KL}(p, q) = \sum_{x \in X} p(x) \log\left(\frac{p(x)}{q(x)}\right) \geq 0.$$